

Two Samples, One Difference

AP Statistics · Topic 5.8 (CED 4.6) · B: 2027-01-22 · E: 2027-03-10 · TEACHER KEY

CED TETHER

- **ENDURING UNDERSTANDING: UNC-3** — Probabilistic reasoning allows us to anticipate patterns in data.
- **Skills: 3.B** — Determine parameters for probability distributions. | 3.C — Describe probability distributions. | 4.B — Interpret statistical calculations and findings to assign meaning or assess a claim.
- UNC-3.T Determine parameters of a sampling distribution for a difference in sample means. [Skill 3.B]
- UNC-3.T.1 For a numerical variable, when randomly sampling from two independent populations with means μ_1 and μ_2 and standard deviations σ_1 and σ_2 , the sampling distribution of the difference in sample means has mean $\mu = \mu_1 - \mu_2$ and standard deviation $\sigma = \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}$. UNC-3.T.2 If sampling without replacement and sample sizes are less than 10% of population sizes, the difference is negligible.
- UNC-3.U Determine whether a sampling distribution of a difference in sample means can be described as approximately normal. [Skill 3.C]
- UNC-3.U.1 The sampling distribution of the difference in sample means can be modeled with a normal distribution if both population distributions can be modeled with normal distributions. UNC-3.U.2 If population distributions cannot be modeled with normal distributions, the sampling distribution can be approximated by a normal distribution if both sample sizes are at least 30.
- UNC-3.V Interpret probabilities and parameters for a sampling distribution for a difference in sample means. [Skill 4.B]
- UNC-3.V.1 Probabilities and parameters for a sampling distribution for a difference of sample means should be interpreted using appropriate units and within the context of specific populations.

Essential question: How can two sample means use different pathways to justify one normal model for their difference?

DOK-3 skill: 4.B · **FRQ pattern:** two-shape-routes-for-a-mean-difference

DOK rationale: Part (c) coordinates a separate shape pathway for each sample mean, judges two incomplete model claims, and uses a small-sample counterfactual to trace when the tail probability loses support.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK: 1**
→ 3 **Minutes: 45**

Teacher does

- Board slide up at the bell. Ask which details govern the sampling model.
- Begin the Topic 5.8 video follow-along at minute 5.
- Return at minute 33. Require separate shape arguments and the counterfactual check.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Name one argument to retain and one that needs more evidence.
- Complete the follow-along, finish (a)–(c), revise, and turn in.

Questions to ask

- Which distribution are we justifying for North: individual waits or sample means?
- Why can the two groups use different routes to approximate normality?
- What changes if the skewed South sample has size 8?
- What repeated-sampling event does 0.0273 describe?

Adult role

Solo teacher. Justify North and South separately, then combine them using independence.

[WARN] WATCH FOR

- Requiring both samples to have size at least 30 even when one population is normal.
- Assuming one normal component automatically fixes an unjustified second component.
- Subtracting variances because the statistic subtracts means.

SCORING PART (c) — E / P / I

Expected elements

- **uses-north-normal-route** (required) — Uses North population normality despite $n_N = 12$
- **uses-south-clt-route** (required) — Uses South $n_S = 48$ with the CLT
- **coordinates-difference-model** (required) — Uses both component models and independence for 0.0273
- **adjudicates-both-accounts** (required) — Diagnoses both incomplete rules
- **traces-counterfactual** (required) — Uses $n_S = 8$ to expose the combined-size shortcut

E	Justifies both components by distinct routes, coordinates independence, diagnoses both accounts, and explains the $n_S = 8$ consequence.
P	Reaches the warranted verdict but omits a route, independence, or the counterfactual consequence.
I	Requires both populations normal or uses one/combined sample size to justify both means.

[STAR] TODAY'S PROBLEM — annotated

Suppose parcel pickup waits have the shown population parameters. North waits are approximately normal; South waits are strongly right-skewed. Independent SRSs are taken without replacement, and the service wants $P(\bar{X}_N - \bar{X}_S > 2.5)$.

Rina: “South is skewed, so I would not use a normal model for the difference.”

Oscar: “North is normal and the combined sample size is 60, so the normal model is valid; the probability is 0.0273.”

Locker samples

Locker	n	μ	σ
North	12	6.0	1.8
South	48	4.8	3.0

FIRST TAKE (5 min)

Identify one argument you would retain and one that needs more evidence. Cite a population shape or sample size.

Key: Not graded. Each account contains a claim that still needs component-level justification.

Rules callout “JUSTIFY EACH SAMPLE MEAN BEFORE THEIR DIFFERENCE (keep on the page)” is printed on the student sheet.

DOK 1 (a) State the mean of $\bar{X}_N - \bar{X}_S$ and translate the event above into context.

Key: The sampling distribution is centered at $\mu_N - \mu_S = 6.0 - 4.8 = 1.2$ minutes. The event says the sample mean North wait exceeds the sample mean South wait by more than 2.5 minutes.

DOK 2 (b) Verify both 10 percent conditions, compute the sampling-distribution SD, and calculate the probability under a normal model.

Key: North: $12 \leq 0.10(600) = 60$; South: $48 \leq 0.10(1,000) = 100$. The SD is $\sqrt{1.8^2/12 + 3.0^2/48} = \sqrt{0.4575} = 0.676387$. Thus $z = (2.5 - 1.2)/0.676387 = 1.92198$ and the upper-tail probability is $0.027304 \approx 0.0273$.

DOK 3 (c) Adjudicate both accounts and decide whether 0.0273 is warranted. Justify each sample mean separately using its shape and sample size, coordinate independence, then use a South sample of 8 to expose the consequence of any rejected shortcut.

Key: The probability is warranted, but both explanations are incomplete. North’s normal population supports $\bar{X}_N$ with $n_N = 12$. For skewed South, $n_S = 48 \geq 30$ supports $\bar{X}_S$ by the CLT. Independence then supports an approximately normal difference. Rina would discard a justified probability by requiring both populations to be normal. Oscar’s combined-size rule fails because each component needs support: with skewed South and $n_S = 8$, its sample mean is not assured normal, so neither is the difference. That shortcut would make readers trust unsupported tails.

EXIT

One thing the video changed about my first take: _____